Amplifying Accessibility in Artificial Music Systems

Ethical Framework and Participatory Design Plan

Prepared for:
Drake Music Northern Ireland Board

Gráinne Meyer
gmeyer01@qub.ac.uk

PhD Research Project
Queen's University Belfast
Sonic Arts Research Centre

Supervised by
Professor Franziska Schroeder
Professor Austen Rainer

Overview

Amplifying Accessibility in Artificial Music Systems is a PhD research collaboration between Queen's University Belfast and Drake Music Northern Ireland. The project develops a gesture-controlled digital musical instrument co-designed with musicians in the *Wired* ensemble, using machine learning to personalise the instrument to each musician's movement style and preferences.

The purpose of this document is to respond to feedback from the Board concerning the project's treatment of movement. Specifically, the feedback raises questions about: the distinction between intentional and unintentional movement and whether this frames involuntary movement negatively; the potential for participants to feel their movements are being evaluated or corrected; the risk that machine learning could 'filter out' disability; and how the design supports participant confidence and autonomy.

This document addresses those potential concerns directly, setting out how the project approaches movement, agency, and machine learning, and presenting the participatory design methodology through which disabled musicians will shape the instrument's development. The guiding principle throughout is that movement is inherently expressive and individual - never something to be corrected, normalised, or filtered out.

Clarifying Key Points Raised by the Board

Concern about the distinction between "intentional" and "unintentional" movement

The Board raised important questions about how the system treats "unintentional movement." In practice, the instrument does not make any distinction between intentional and unintentional movement at all. The system does not analyse, label, interpret, or filter movement in these terms. Instead, the musician simply selects which of their movements they would like to use for making sound. Any movements they do not choose as part of their musical vocabulary, whether those are spontaneous, habitual, or the user simply does not wish to make music with, are treated neutrally by the system.

In other words, the concept of "unintentional movement" is not something this technology evaluates. The system responds only to the gestures the musician deliberately selects during calibration and does not make assumptions about the meaning, value, or intention of any other movement.

Concern that participants may feel their movement is being judged or evaluated

The Board noted that participants might feel their movement is being assessed or judged. The instrument does not evaluate or critique any form of movement. During calibration, musicians choose which gestures they want to use to create sound, and the system learns only those gestures. There is no underlying model of "correct" movement and no interpretation of how a participant ought to move.

Concern that machine learning could "filter out" disability or erase disabled embodiment

The Board raised concerns that machine learning systems sometimes impose normative assumptions about how bodies should move. This instrument **does not** use machine learning in that way. It does not contain any generalised model of typical or ideal movement, and it does not correct, normalise, or suppress any bodily pattern.
The machine learning component functions solely to recognise and respond to the specific gestures each musician *personally* selects during calibration. Every model is unique to the individual and reflects only their preferred expressive vocabulary. Movements that are not part of this chosen vocabulary are **not** filtered or modified; they are simply not mapped to sound, and this decision lies solely with the user. Disability-related movement is therefore never interpreted as error or noise, nor is it removed or altered in any way.

PART 1

Movement, Agency, and Machine Learning

The instrument

The instrument is a browser-based digital musical instrument that allows musicians to create and manipulate sound through body movement tracked via a standard webcam. When a musician moves, the system detects their position and gestures using pose-estimation and translates these into musical output in real time. The result is an instrument that can be played without requiring the musician to hold, press, or physically manipulate any object, opening possibilities for musicians whose bodies interact with traditional instruments in limited or fatiguing ways.

A key design consideration is how the system handles the distinction between movements a musician wishes to use musically and movements they do not wish to use for making music. This distinction is made by the participant themselves, through a creative calibration process, demonstrating the gestures they want to use. The system learns to recognise those gestures. It does not classify movements as good or bad, nor does it decide on the musician's behalf which movements should or should not make sound.

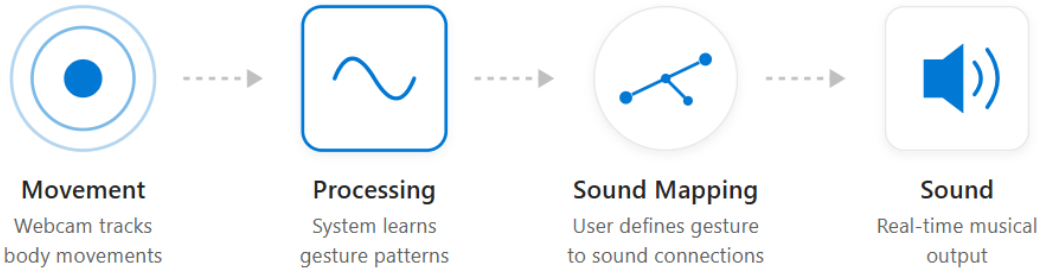


Figure 1: System Overview

Creative calibration

The machine learning component of this project is defined as a creative calibration tool that operates entirely under the user's control. Its function is to learn the distinction between movements that a particular musician wishes to use musically and movements that same musician prefers not to use. This is a distinction that only the musician themselves can define.

Personalisation

- User-led gesture selection: During a *creative calibration* phase, each musician decides and demonstrates the gestures they wish to use for making music.
- Personal model training: The system learns from these demonstrations to recognise the participant's chosen gesture vocabulary. This creates a personalised model specific to that individual, not a generalised model of 'correct' movement.
- Transparent adjustment: Participants can see when the system is responding to their movements and can adjust their calibration at any time.
- Reversible choices: All calibration decisions are reversible. A musician might choose to add a gesture to their sound-producing vocabulary, remove one, or start fresh with a completely new configuration.

This process is designed as *creative customisation* rather than *corrective filtering*. The question posed to participants is not 'which of your movements need to be filtered out?' but rather 'which of your movements would you like to make music with?' This reframing centres participant agency and artistic intent.

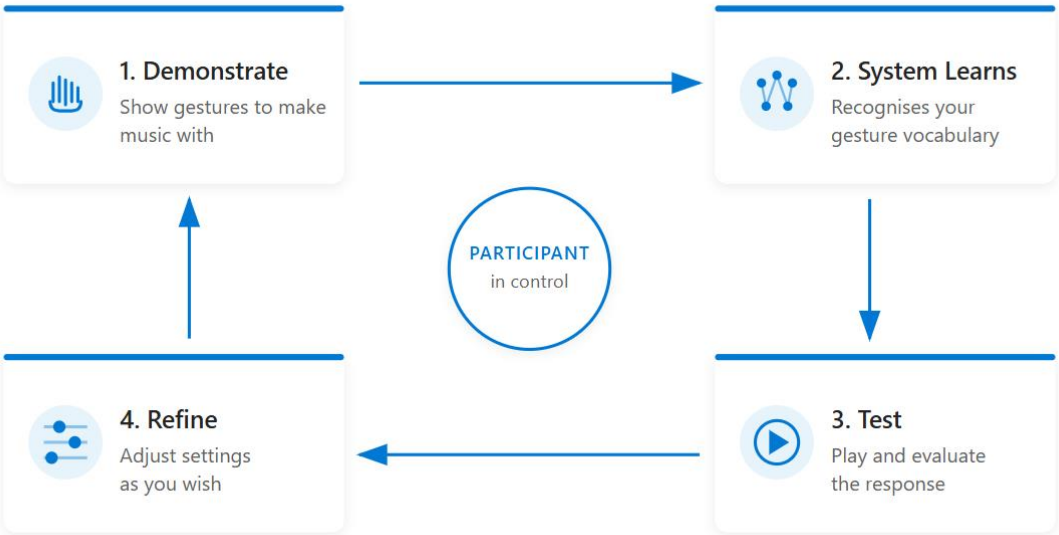


Figure 2: The creative calibration process

Core position on movement

The project is built on the premise that all movement is expressive, individual, and positive. Every gesture carries creative potential; no two musicians move alike; and movement is a source of agency and self-expression. These principles directly inform how the instrument is designed, how calibration works, and how participants experience the technology. participants choose which movements make music because they are the only ones who can. No system can determine from the outside which gestures feel meaningful or comfortable to a particular person. That knowledge belongs to the musician, and so does the choice.

What the system does not do

To address concerns directly, it is important to acknowledge explicitly that the system does **not**:

- ✗ Categorise movement as 'good' or 'bad'
- ✗ Label movement as 'correct' or 'incorrect'
- ✗ Classify movement as 'typical' or 'atypical'
- ✗ Autonomously filter out disability
- ✗ Correct, normalise, or remediate movement

This explicit rejection of normative body models distinguishes our approach from many conventional motion-tracking systems, which often embed assumptions about 'standard' movement patterns. This design instead follows the principle that the musician defines what is meaningful for them.

Transparency and trust

Participants should always understand what the instrument is doing and why. If a system makes hidden corrections or autonomous decisions, participants cannot truly control their experience. The design therefore ensures transparency throughout the process:

- Visual feedback: The interface clearly shows when movements are being detected and when they are producing sound, so participants always know what the system is responding to.
- Accessible calibration: The calibration process is designed to be understandable and accessible, using plain language and avoiding technical jargon.
- Participant documentation**:** Musicians receive clear and straightforward information about how their calibration choices shape the system's responses, with no hidden processes.

Commitment to agency

The instrument is designed around the following commitments:

Agency	Participants make all meaningful decisions about how the instrument responds to their movements
Empowerment	The technology exists to amplify participants' musical expression and creative intent, not to make assumptions, shape or correct it
Autonomy	Participants retain full control over their calibration choices
Confidence	Transparent, predictable behaviour supports participants in developing musical control
Trust	Collaborative design and transparent system behaviour build trust

Summary

This project develops a browser-based digital musical instrument that allows musicians to create and shape sound using their own movement, through personalised gesture selection and mapping. Each musician chooses which of their movements they wish to use for musical control, and the system responds only to those chosen gestures.

Machine learning is used as a customisation tool to support this personalisation. It learns only the gestures that a musician deliberately selects during calibration, enabling the instrument to adapt to each individual's expressive style. The system does not apply generalised movement models or attempt to shape, correct, or evaluate how anyone moves; its role is to support the musician's own creative decision-making.

The design of the instrument prioritises clarity, predictability, and ease of use. Visual feedback makes the system's behaviour transparent, calibration choices are reversible, and the overall experience is structured to support confidence, comfort, and creative exploration.

The participatory design process places musicians in the Wired ensemble at the centre of development. Through collaborative workshops, participants explore the prototype, shape their gesture vocabularies, influence sound design, and guide iterative improvements. Each development cycle responds directly to their insights, ensuring that the instrument evolves in line with their needs, preferences and creative wishes.

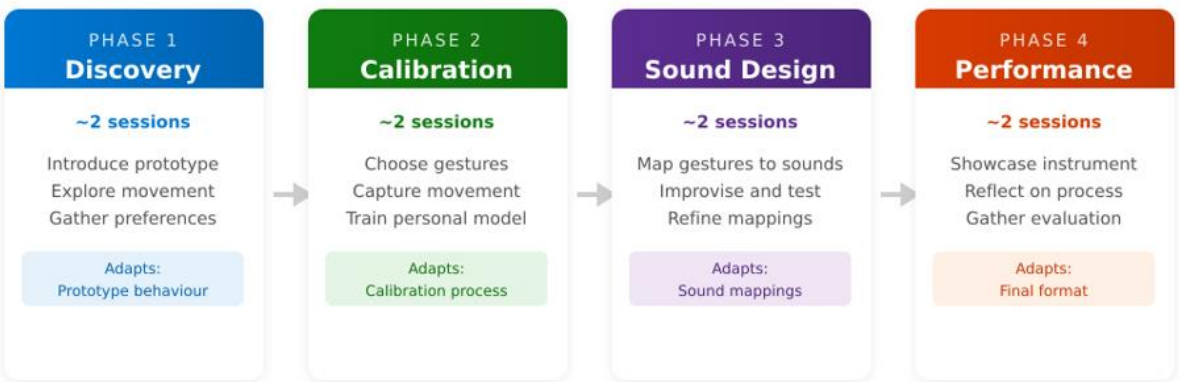
The project aims to create an instrument that amplifies the musician's creative intentions, supports accessible music-making, and develops new opportunities for expression through technology.

PART 2

Participatory Design Workshop Plan

Overview

This workshop plan outlines a participatory design programme working with musicians in the Wired ensemble as co-designers. The programme follows an agile, iterative approach where activities and directions emerge through collaboration with participants rather than being entirely predetermined.

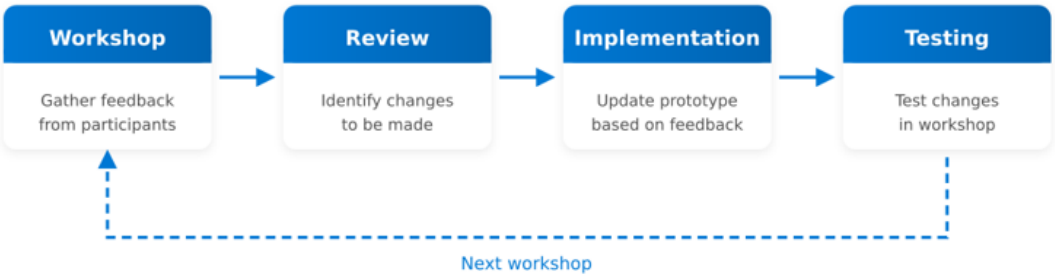


Duration	Approximately 10 weeks (space for additional sessions in each phase if required)
Phases	Four phases with flexible session counts
Session length	Approximately one hour per session
Approach	Agile and participant-led

Iterative development

The workshop programme uses an agile methodology where participant feedback directly shapes development between sessions. This is not a linear process with fixed outcomes, but a responsive cycle where the instrument evolves based on what participants discover, prefer, and create.

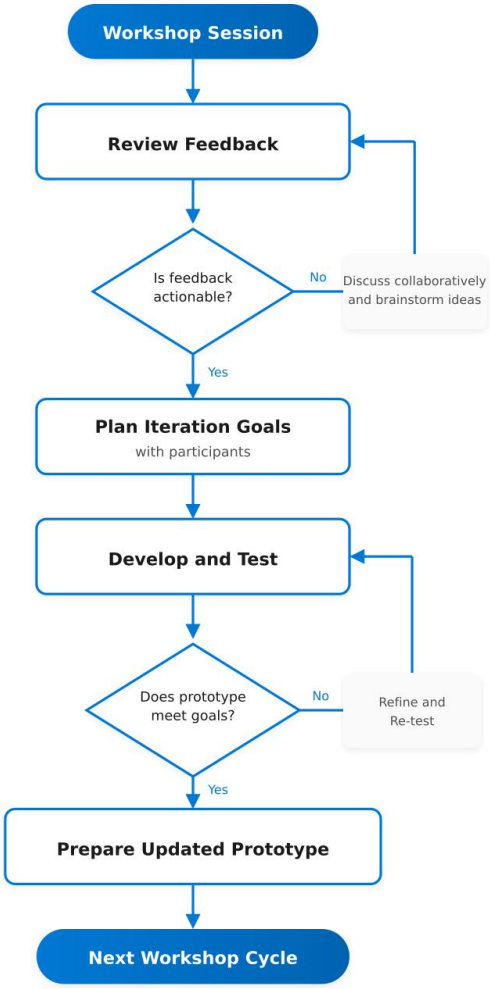
Between workshops, feedback and input from the Wired ensemble is implemented in the design and technical updates are made in the code. A clear record links each suggestion to the resulting change. At subsequent sessions, participants see exactly how their input has been implemented in development and test these changes and decide if they are happy with the outcome or if further development or changes are required.



Workshop Process

Each workshop follows an agile process that ensures participant feedback is captured, reviewed, and implemented systematically.

While this document outlines a structure of phases and workshops, the specific activities within each session will be defined collaboratively and adjusted based on what emerges. A rigid, predetermined plan would contradict the participatory ethos. Instead, each workshop responds to what participants discovered in the previous session, what questions arose, and where they wish to focus their creative energy.



Programme phases

The number of sessions within each phase is determined collaboratively based on participant needs and progress.

Phase 1: Discovery

Purpose	Introduce prototype instrument, explore movement as creative medium
Activities	Free play and exploration; identify what works and needs improvement
Insights	Movement ranges, sound preferences, technical limitations

Phase 2: Calibration

Purpose	Build personalised movement vocabulary through creative calibration
Activities	Choose gestures for music making; capture data, train personal model
Insights	Gesture vocabularies, calibration time, tracking reliability

Phase 3: Sound Design

Purpose	Map gestures to sounds, explore musical possibilities
Activities	Short improvisations to test expressiveness of mappings
Insights	Mapping preferences, expressiveness feedback, future needs

Phase 4: Performance

Purpose	Showcase instrument, reflect on process, gather evaluation
Activities	Performance or demonstration; reflection and evaluation discussion
Insights	Experience feedback, process documentation, future directions

Complete Workshop Reference

PHASE 1

Discovery

Agile: ~2 sessions

Purpose

Introduce prototype instrument, explore movement as creative medium

What Happens

- Free play and exploration of prototype
- Identify what works and needs improvement

Insights Gathered

- Comfortable movement ranges
- Sound preferences
- Technical limitations observed

Prototype adapts to observed movement and comfort levels.

PHASE 2

Calibration

Agile: ~2 sessions

Purpose

Build a personalised movement vocabulary through creative calibration

What Happens

- Choose gestures for music making
- Capture data, train personal model

Insights Gathered

- Personal gesture vocabularies
- Calibration time per participant
- Real time tracking reliability

Calibration adapts to each participant's movement style.

PHASE 3

Sound Design and Play

Agile: ~2 sessions

Purpose

Map gestures to sounds, explore musical possibilities

What Happens

- Short improvisations to test how expressive or restrictive mappings feel

Insights Gathered

- Sound mapping preferences
- Expressiveness and fatigue feedback
- Future development needs

Sound mappings refined for each participant's musical interests.

PHASE 4

Review and Performance

Agile: ~2 sessions

Purpose

Showcase the instrument, reflect on the co-design process, gather evaluation

What Happens

- Performance or demonstration
- Reflection and evaluation discussion

Insights Gathered

- Overall experience feedback
- Documentation of the process
- Future development directions

Final format shaped by what participants wish to share.

Next steps

The participatory design approach means that the specific activities and directions will continue to evolve based on what emerges through collaboration with the Wired ensemble.